DNA Replication 1

10/26/26

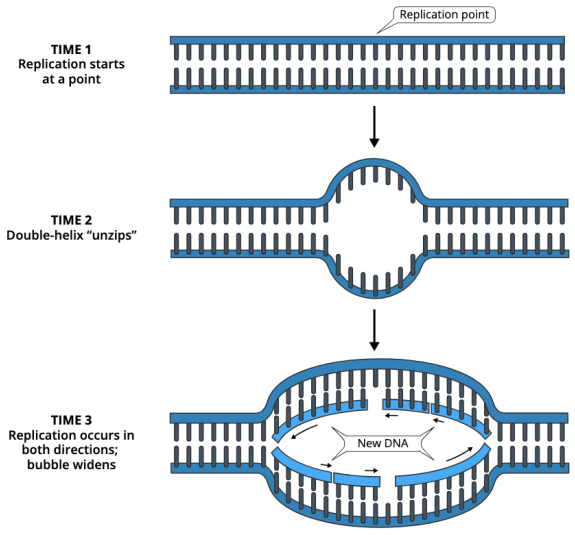
DNA needs to be stable, copy-able and contain information. DNA strands are anti-parallel, new strands are synthesized in the 5’ to 3’ direction and the complementary strands run in the opposite 3’ to 5’ direction.

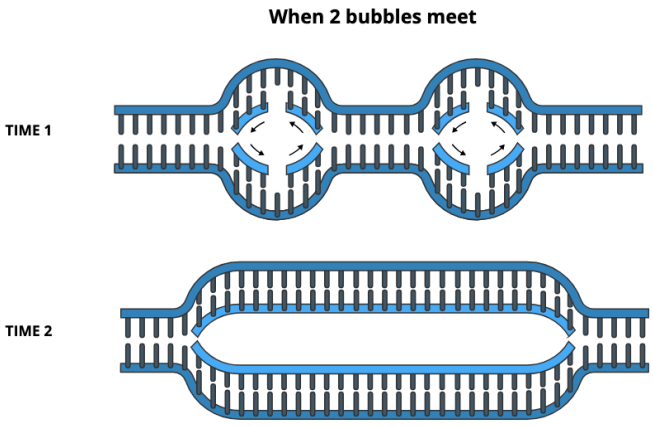
For DNA synthesis to occur, enzymes need to catalyze formation of phosphodiester linkages between deoxyribonucleotides that are complementary to a template strand.

Rules to synthesizing a new complementary strand of DNA

1. DNA synthesis occurs only in the 5’ to 3’ direction, meaning new bases are being added to the 3’ end of an existing strand
2. DNA polymerase (enzyme that builds new DNA strands) can only add new bases to the 3’ end of an existing complementary strand. It can’t synthesize a complementary strand of DNA from scratch and the enzyme can only work if the new complementary strand has already been started. The enzyme can add deoxyribonucleotides to the ribonucleotides that are hydrogen bonded to a complementary DNA strand
3. When DNA replication occurs, gaps form between the ends of newly synthesized segments (so the gap in the sugar phosphate backbone between two adjacent nucleotides). DNA ligase closes this gap in the sugar phosphate backbone without adding new bases to the 3’ of the growing strand

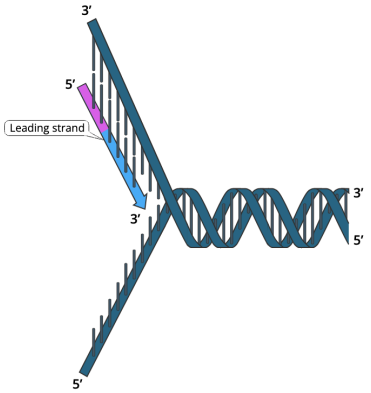
The specific points on a DNA strand where replication starts is known as the origin of replication. Replication occurs in both directions and therefore widens the bubble as both sides of the strand are being replicated.





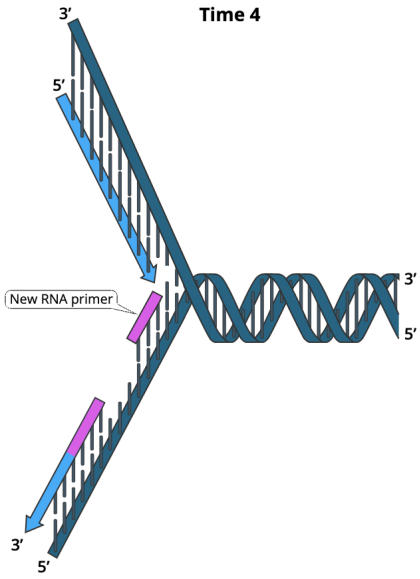
Replication Fork

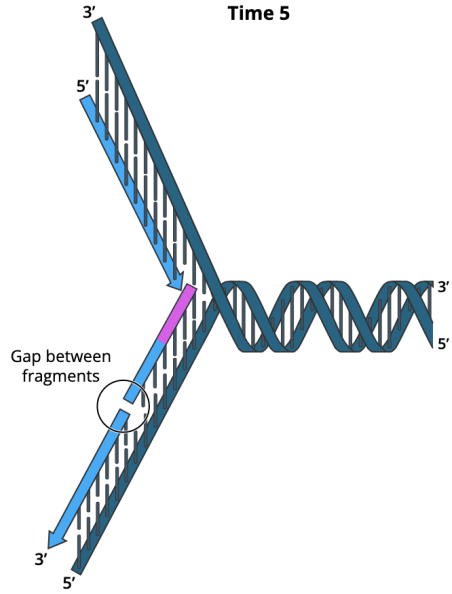
- This is where DNA replication starts to happen
- In the picture, the top strand runs from 3’ to 5’, so DNA polymerase can use it as a template to synthesize a complementary strand in the 5’ to 3’ direction
    - Before this happens, primase synthesizes a short sequence of RNA that is complementary to the DNA on the top strand. The new result is RNA primer because it prepares (primes) the strand for DNA polymerase
    - Unlike DNA polymerase, primase can start synthesis from scratch on a single stranded template strand
- One the primer is made, DNA polymerase can start adding deoxyribonucleotides
    - The strand that is synthesized toward the replication fork is called the leading strand



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- DNA polymerase adds deoxyribonucleotides to create the lagging strand because its synthesis lags behind the synthesis of the leading strand
- DNA synthesis continues and moves away from the fork, which continues to unwind in the opposite direction, creating a region of single-stranded DNA that has opened up behind the first new segment of the lagging strand. In response, primase synthesizes another short RNA primer in the 5’ to 3’ direction on the lagging strand. When DNA polymerase is adding a new fragment to the lagging strand, it stops when it gets to a previously synthesized fragment, leaving a gap.





Ligase then catalyzes formation of a phosphodiester bond that closes the gap, creating a continuous sequence. A specialized version of DNA polymerase comes in to remove the RNA primer and replaces it with deoxyribonucleotides. The RNA primer disappears as DNA as now in its place (since chromosomes are made of DNA, not RNA). The specialized enzyme is called DNA polymerase I or just pol I in bacteria, the main enzyme is DNA polymerase III or pol III.

All of the different enzymes and proteins involved have combined to form the replisome. It consists of…

- ssDNA - single stranded DNA
- dsDNA - double stranded DNA
- Place of split for the formation of the replication fork
- Helicase, which unzips the DNA by breaking hydrogen bonds between complementary pairs to create ssDNA
- Thioredoxin molecules that move along DNA strands as replication continues
- Topoisomerase, which breaks and rejoins DNA ahead of the replisome to relieve the twisting force created by helicase

DNA polymerase: An enzyme that catalyzes the formation of phosphodiester linkages between deoxyribonucleotides, forming the primary structure of DNA.

DNA ligase: An enzyme that catalyzes the formation of phosphodiester linkages in gaps between two existing nucleotides in double-stranded DNA.

Origin of replication: A location (a point) where enzymes open the DNA double helix so replication can begin.

Replication fork: The y-shaped structure visible in micrographs of DNA during replication. This is the site where the replisome works, synthesizing the leading and lagging strands.

Primase: An enzyme that adds a short RNA primer to single-stranded DNA, so DNA polymerase can start adding complementary bases.

Replisome: The multi-molecular machine that works at the replication fork, copying DNA.

Helicase: An enzyme in the replisome that opens the DNA double helix, creating single-stranded template DNA for leading and lagging strand synthesis.

Topoisomerase: An enzyme that is important in DNA replication. It breaks and rejoins DNA to relieve the tension created when helicase continuously opens the double helix.